

# AI Content Detection

## Business Model & Infrastructure Guide

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Bangla-first multimodal misinformation detection platform

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## 1. Product Stack & Feature Overview

The platform is organised into three independent pillars. Each pillar operates autonomously — a user may invoke any one without triggering the others.

### Pillar 1 — Independent AI Detectors

| Module                                | Model                       | Speed          | VRAM    | Accuracy |
|---------------------------------------|-----------------------------|----------------|---------|----------|
| AI Text Detection                     | FastGPT + mDeBERTa (Bangla) | 5 texts / sec  | 5 GB    | F1: 88%  |
| AI Image Detection (high accuracy)    | FSD CVPR-2025               | 13 sec / image | 10.9 GB | F1: 88%  |
| AI Image Detection (fast / free tier) | FFreq                       | 4 images / sec | 1.1 GB  | —        |
| AI Video Detection                    | VideoVeritas (2026)         | TBD (batch)    | 30 GB   | TBD      |

■ VideoVeritas is a 2026 model; accuracy benchmarks are pending. It should be offered as a beta feature at launch with a clearly communicated SLA.

### Pillar 2 — Fact-Check Pipeline (Claim Extract → Verify)

Claims are first extracted from the submitted content, then passed to the fact-verification engine. This is a sequential two-step pipeline, not two independent features.

#### Step 1 — Claim Extraction

| Input type        | Pipeline                            | Cost / call         | VRAM required                  |
|-------------------|-------------------------------------|---------------------|--------------------------------|
| Image + text      | Mahdi's pipeline                    | ~\$0.0005           | Pool A (shared)                |
| Video — Pro tier  | Qwen2.5-flash (API)                 | ~\$0.001 / 30s clip | No GPU (API)                   |
| Video — Free tier | wav2vec2 ASR + LLaVA-Video-Qwen2-VL | \$0.003–0.005       | wav2vec2 ~1 GB<br>LLaVA ~16 GB |

#### Step 2 — Fact Verification

Extracted claims are passed to OpenFactVerification (2025) which queries Serper API for evidence retrieval and uses cheap OpenRouter / Qwen models for reasoning. No GPU is required.

| Metric                     | Value                         |
|----------------------------|-------------------------------|
| Throughput                 | ~6 checks / min (10 sec each) |
| Cost per claim             | ~\$0.0017 (\$1 ≈ 600 claims)  |
| GPU requirement            | None — pure API cost          |
| Average claims per article | ~5 → ~\$0.0085 / article      |

### Pillar 3 — Content Filtering (Feed Control)

Qwen3-7B handles all content filtering — political framing detection (via the CUET solution), misinformation context analysis, and AI-generated content flagging. No visual-only classifier is used in this pillar.

| Model    | Role                                                                    | VRAM      | Speed    | Pool               |
|----------|-------------------------------------------------------------------------|-----------|----------|--------------------|
| Qwen3-7B | Semantic filter — political framing, misinformation context, AI content | ~14–16 GB | Moderate | Pool B (on-demand) |

Active filter categories: Political (CUET solution), AI-generated content, Misinformation. Vulgar content filtering planned for a later release.

## 2. GPU Infrastructure Strategy

Never run everything simultaneously. GPU costs are highly uneven across modules. The core strategy is three pools with distinct availability profiles.

### Pool A — Always On

|                   |         |
|-------------------|---------|
| Text detection    | 5 GB    |
| FFreq image model | 1.1 GB  |
| Total             | ~6.1 GB |

RTX 3090 (24 GB) or RunPod equivalent · ~\$0.15/hr · ~\$110/month

### Pool B — On-Demand

|                         |                                                 |
|-------------------------|-------------------------------------------------|
| FSD image model         | 10.9 GB                                         |
| Qwen3-7B text filter    | ~16 GB                                          |
| LLaVA-Video (free tier) | 16 GB                                           |
| Note                    | Models loaded sequentially — not simultaneously |

RTX 4090 (24 GB) or 2× RTX 3090 · ~\$0.35–0.50/hr · ~\$80–150/month

### Pool C — Job Queue Only

|                |                              |
|----------------|------------------------------|
| VideoVeritas   | 30 GB                        |
| Queue strategy | Spin up only when jobs exist |
| SLA            | Results within 30 minutes    |
| Hardware       | A100 40 GB on Vast.ai        |

A100 40 GB · ~\$1.50/hr · ~\$30–60/month at launch volumes

■ Pool B scheduling conflict: Qwen3-7B (16 GB) and LLaVA-Video (16 GB) cannot co-exist on a single 24 GB card. Use model-swapping with a job queue (Redis + Celery recommended) or provision two separate Pool B instances once traffic justifies it.

### Total Monthly Infrastructure Cost

~\$110

Pool A (always on)

~\$80–150

Pool B (on-demand)

~\$30–60

Pool C (job queue)

~\$220–320

Total estimate

### 3. Cost-Per-Request Analysis

GPU costs are amortised: hourly rental cost divided by maximum request throughput per hour. API costs (fact-check pipeline) are variable per call.

#### Pillar 1 — AI Detectors

| Module               | GPU Pool         | \$/hr  | Req/hr | Cost / request |
|----------------------|------------------|--------|--------|----------------|
| Text detection       | Pool A (shared)  | \$0.15 | 18,000 | ~\$0.000008    |
| Image — FFreq        | Pool A (shared)  | \$0.15 | 14,400 | ~\$0.00001     |
| Image — FSD          | Pool B on-demand | \$0.50 | 276    | ~\$0.0018      |
| Video — VideoVeritas | Pool C A100      | \$1.50 | 20–40  | ~\$0.04–0.08   |

#### Pillar 2 — Fact-Check Pipeline

| Step                         | Route                               | Cost / call    |
|------------------------------|-------------------------------------|----------------|
| Claim extract — image+text   | Mahdi's pipeline (Pool A amortised) | ~\$0.0005      |
| Claim extract — video (Pro)  | Qwen2.5-flash API                   | ~\$0.001 / 30s |
| Claim extract — video (Free) | wav2vec2 + LLaVA Pool B GPU time    | ~\$0.003–0.005 |
| Fact verification per claim  | Serper + OpenRouter/Qwen            | ~\$0.0017      |
| Full pipeline (avg 5 claims) | Pro route                           | ~\$0.012       |
| Full pipeline (avg 5 claims) | Free route (with video)             | ~\$0.025       |

#### Pillar 3 — Content Filtering

| Module               | Notes                    | Cost / item    |
|----------------------|--------------------------|----------------|
| Qwen3-7B             | Pool B on-demand, ~16 GB | ~\$0.001–0.002 |
| Filter pass per item | Qwen3-7B only            | ~\$0.001–0.002 |

*Key insight: The free tier's 'free' LLaVA pipeline for video claim extraction costs more per request than the Pro Qwen API route due to Pool B GPU time. Free for the user does not mean cheap for you. Video claim extraction should therefore be gated at Pro tier only.*

## 4. Tier Model — Free, Pro & Enterprise

### Free Tier — "Verify"

Goal: acquisition. Keep it genuinely useful but cap every expensive operation. Video claim extraction is excluded entirely — LLaVA GPU cost makes it unviable at scale on the free tier.

| Feature                  | Limit            | Model used                              |
|--------------------------|------------------|-----------------------------------------|
| AI text detection        | 20 checks / day  | mDeBERTa (Pool A)                       |
| AI image detection       | 10 images / day  | FFreq only (Pool A)                     |
| AI video detection       | Not available    | —                                       |
| Fact-check pipeline      | 3 articles / day | Mahdi + OpenFactVerify (no video input) |
| Claim extraction — video | Not available    | LLaVA cost prohibitive                  |
| Content filtering        | Not available    | Qwen3-7B on Pool B — reserved for Pro   |
| Feed control             | 1 source / feed  | Political + AI categories               |
| API access               | Not available    | —                                       |

Estimated cost to serve per active free user/month: ~■50–100 (~\$0.50–1.00)

### Pro Tier — ■699 / month (~\$6.50 USD)

Target: journalists, researchers, content moderators, students, fact-checking organisations.

| Feature                  | Limit                      | Model used                           |
|--------------------------|----------------------------|--------------------------------------|
| AI text detection        | 1,000 / day                | mDeBERTa (Pool A)                    |
| AI image detection       | 200/day FFreq + 20/day FSD | FFreq (Pool A) / FSD (Pool B)        |
| AI video detection       | 10 videos / day            | VideoVeritas — queued, 15–30 min SLA |
| Fact-check pipeline      | 100 articles / month       | Qwen2.5-flash + OpenFactVerify       |
| Claim extraction — video | Included                   | Qwen2.5-flash (~\$0.001/30s)         |
| Content filtering        | Full                       | Qwen3-7B (Pool B)                    |
| Feed control             | Unlimited sources          | All 3 active categories              |
| API access               | 5,000 calls / month        | All endpoints                        |
| Priority queue           | Yes                        | Pool B jobs prioritised              |

Estimated cost to serve per active Pro user/month: ~\$2.50–3.50. Gross margin at ■699: ~50–60%.

### Enterprise / Newsroom Tier — ■8,000–25,000 / month

Target: media houses, fact-checking NGOs, government digital literacy programs, universities, social platforms.

- Unlimited text + image detection (fair-use SLA)

- Dedicated video queue — 5-minute SLA for VideoVeritas
- Dedicated Pool B instance (no model-swap delay for Qwen3-7B)
- Custom fact-check volume — monthly report exports included
- White-label API with custom domain and branding
- On-premises Docker deployment option (client provides GPUs)
- Dedicated support channel + onboarding assistance
- Custom feed control rules and category tuning



5. Revenue Projections (Month 6)

80 users  
Pro subscribers

2 orgs  
Enterprise orgs

est.  
API (pay-as-you-go)

~63%  
Net margin

| Revenue stream                  | Volume   | Unit price  | Monthly (BDT) |
|---------------------------------|----------|-------------|---------------|
| Pro subscribers                 | 80 users | ■699        | ■55,920       |
| Enterprise contracts            | 2 orgs   | ■12,000 avg | ■24,000       |
| API pay-as-you-go               | —        | Variable    | ■8,000        |
| Gross revenue                   |          |             | ■87,920       |
| Infrastructure (Pool A+B+C)     |          |             | –■25,000      |
| API costs (Serper + OpenRouter) |          |             | –■8,000       |
| Net revenue                     |          |             | ~■54,920      |

At ■54,920/month net (~\$500 USD), this is a self-sustaining product covering all infrastructure costs and providing meaningful income as a student-run project. A single additional enterprise contract adds ■8,000–25,000 directly to net margin.

## 6. Launch Sequence

### Month 1–2 · Free Tier Launch

- Pool A only — text detection + FFreq image.
- No video, no Qwen3-7B, no LLaVA. Infrastructure: ~\$110/month.
- Goal: 200 free users. Validate detection quality and UX.
- Key action: Publish Bangla AI text detection benchmarks publicly — this is your differentiator and will attract academic/media attention.

### Month 3–4 · Pro Tier Launch

- Bring Pool B online. Enable FSD image, Qwen3-7B filtering, Qwen2.5-flash claim extraction.
- VideoVeritas beta — Pro-only, 30-minute batch SLA.
- Goal: 30 paying Pro users = ■20,970/month. Infrastructure ~\$220/month.
- You are now cash-positive.

### Month 5–6 · Enterprise Outreach

- Approach Prothom Alo, Daily Star, bdnews24 for enterprise pilots.
- Contact fact-checking NGOs and digital literacy programs.
- Offer 30-day free enterprise trials to 3–5 newsrooms.
- One signed enterprise contract at ■12,000 covers all infrastructure.

### Month 6+ · API & Developer Ecosystem

- Launch pay-as-you-go API with 1,000 free credits for developers.
- Developers build on top of your platform — compounding user growth.
- Consider open-sourcing the Bangla text detection model — drives credibility and inbound enterprise leads.

## 7. Strategic Notes

### Lead with Bangla

Your mDeBERTa-based Bangla AI text detector is the only production-quality model of its kind. No international product has this. Market aggressively to Bangladeshi media, government digital literacy programs, and the NLP research community. This is your moat.

### Video detection is premium bait

The 30 GB VRAM requirement makes video detection inherently scarce. Use that scarcity. 'First Bangla-context AI video detection service' is a real, defensible headline. Never give it away for free.

### Fact-checking drives stickiness

Once a newsroom integrates the fact-check API into their editorial workflow, churn approaches zero. Prioritise newsroom enterprise sales over individual Pro subscriptions in months 3–6.

### The LLaVA free-tier trap

The 'free' wav2vec2 + LLaVA pipeline costs more per request than Qwen2.5-flash due to Pool B GPU time. Do not offer video claim extraction on the free tier. The clean boundary is: free tier = text and image content only.

### Job queue engineering (critical path)

The on-demand Pool B model works only if you have a reliable job queue. Redis + Celery is the recommended stack. Build this before launching Pro — model-swap delays without a queue will result in timeouts and user churn.

### Pricing in BDT vs USD

Price the consumer/Pro tier in BDT (₳699) for local accessibility. Price the API and enterprise tier in USD (\$6 / custom) to signal international readiness and simplify foreign billing.

Document prepared June 2026. Infrastructure costs are estimates based on RunPod / Vast.ai pricing as of mid-2026 and will vary. VideoVeritas (2026) accuracy benchmarks are pending official release.